

Employee Turnover Prediction

Final Report: Data Preparation, Modeling Results, and Interpretability

Prepared for advisor review

June 25, 2026

Abstract

This report summarizes the final employee-turnover prediction pipeline developed from three HR data files. The final design uses file1 and file2 for model development and validation, while file3 is held out as an external test set. The supplied `leave_ind` field is used as the prediction target. Aziva exit-code metadata was audited for label consistency, but excluded from modeling because it would leak outcome information. The selected model, `XGBoost__learned_imputation`, achieved validation PR-AUC 0.2052 and external file3 PR-AUC 0.1462. Since file3 prevalence is 4.87%, this external PR-AUC is approximately 3.00 times the baseline positive rate. The model is therefore useful primarily as a ranking and prioritization tool, not as an autonomous decision system.

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1 Executive Summary

The goal of this project is to predict employee turnover, represented by the supplied `leave_ind` target, and to explain the main risk drivers in a way that can support HR analysis at company, department, and individual employee levels. The final pipeline trains on file1 and file2, validates with employee-grouped validation, and evaluates once on file3 as an external test set.

Bottom line. The selected model is `XGBoost__learned_imputation`. It achieved validation PR-AUC **0.2052**, validation ROC-AUC **0.7615**, file3 PR-AUC **0.1462**, and file3 ROC-AUC **0.7758**. The file3 positive rate is only **4.87%**; therefore, the file3 PR-AUC is approximately **3.00x** the baseline prevalence. In the top 20% highest-risk file3 rows, the model captured **51.63%** of positives at **12.58%** precision.

Key conclusions:

- The model has useful ranking power, especially for prioritizing a subset of employees for review.
- Absolute precision remains modest because turnover is a rare event in file3.
- Age, tenure, and data-timing variables are strong descriptive predictors.
- Actionable or partially actionable signals remain important: compensation, salary trajectory, workload, illness/absence patterns, contract type, and role/manager context.
- The main methodological concerns are source shift between file1/file2 and file3, noisy file2 payment imputation, low event prevalence, and small subgroup sizes.

2 Data Description and Study Design

The project uses three HR Excel data sources. File1 and file2 serve as development data. File3 is reserved as an external test set and is not used for fitting, imputation training, model selection, or threshold selection. The modeling unit is an employee-period row rather than only a single latest employee snapshot. This preserves temporal information, but it also means that interpretation should be framed as ranking employee-period risk rather than predicting one permanent label per employee.

Table 1: Final train/validation/test split summary.

Split	Rows	Employees	Positive rows	Positive rate
Inner train	10,400	3,104	831	7.99%
Validation	2,654	777	207	7.80%
File3 external test	3,140	790	153	4.87%

The lower prevalence in file3 is important. Precision-recall metrics are prevalence-sensitive: a model can retain reasonable ROC-AUC while still producing lower PR-AUC and lower precision when positives are rarer. Therefore, the file3 PR-AUC should be interpreted relative to the 4.87% baseline.

3 Data Quality Issues and Corrective Methods

3.1 Target definition and leakage controls

The supplied `leave_ind` field is used as the target. A specific issue arose around Aziva codes: in file1 and file2, Aziva code 41 corresponds to the resignation concept associated with `leave_ind = 1`; in file3, Aziva code 42 corresponds to the same conceptual outcome. Thus, Aziva code 41 in files 1 and 2 is equivalent to Aziva code 42 in file3 for the resignation label.

However, Aziva codes were not used as predictors. They are exit/outcome metadata and would leak post-event information into the model. They were used only to audit target consistency.

Leakage controls:

- Dropped outcome metadata: `aziva_kod`, `aziva_date`, `aziva_year`, target-like fields, and `leave_ind`.
- Dropped identifiers and bookkeeping fields from model inputs: `source_employee_id`, `fictive_employee`, `calc_month`, `source`, and `year_date`.
- Kept `calc_month` only for chronological ordering and within-period permutation analysis; it is not directly modeled.

3.2 Invalid records and cleaning

Table 2: Invalid-record cleaning rules.

Cleaning rule	Total removed	file1	file2	file3
Negative average illness	14	0	9	5
Non-positive average payment	136	0	136	0
Non-positive average workload	4	0	3	1
Age below 18	11	0	10	1
Negative tenure	0	0	0	0

Overall, 165 of 16,359 rows were removed, leaving 16,194 usable rows. Only six removed rows were positive. Employees above age 75 were intentionally retained because this is feasible in the domain; age below 18 was treated as invalid.

3.3 Payment missingness and imputation

Payment missingness was a major issue, especially in file2. The final pipeline uses a fold-safe similar-employee imputer for payment-related variables. The imputer predicts `avg_Payment`, `Median_Payment`, `stdevp_Payment`, and `salary_change`. It is fitted only on training data inside each validation fold and never on file3. Missingness indicators are added so that the model can learn that the value was originally missing.

Table 3: Cross-validated payment imputation diagnostics.

Source	Imputed field	MAE	RMSE	Median absolute error
file1	avg_Payment	1,776.4	3,809.4	523.6
file1	Median_Payment	1,844.2	3,978.8	544.1
file1	salary_change	371.4	844.0	171.4
file2	avg_Payment	3,672.0	5,123.0	2,738.1
file2	Median_Payment	3,758.3	5,235.8	2,810.3
file2	salary_change	603.9	1,272.1	335.1
overall	avg_Payment	2,364.8	4,271.5	890.4
overall	Median_Payment	2,441.0	4,429.2	900.2
overall	salary_change	447.1	1,018.6	215.7

The imputation diagnostics show that file2 payment imputation is materially noisier than file1. This motivated explicit comparison of learned-imputation, no-payment, and native-missing modeling strategies.

4 Feature Engineering and Preprocessing

The final preprocessing pipeline combines raw HR variables with time-safe historical summaries. Historical features use only current and prior rows for the same employee. This avoids future leakage while allowing features such as previous payment, previous illness, deltas from previous values, prior means, prior standard deviations, slopes per month, and observed-fraction-to-date variables.

Important preprocessing choices:

- Categorical fields are imputed with a missing category and one-hot encoded inside the fitted pipeline.
- Numeric fields are median-imputed when required; an XGBoost native-missing variant was also tested.
- Payment imputation is inside the model pipeline, so validation folds do not share imputation state.
- Protected or unstable descriptive fields such as gender and marital status were excluded from the final default feature list.
- `Tafkid` text was excluded because EDA suggested source instability. Related role/organization codes were interpreted cautiously as contextual predictors, not causal levers.

5 Modeling Strategy

The final model search compared Logistic Regression, Random Forest, and XGBoost under several payment strategies. Class imbalance was handled only through native estimator mechanisms: `class_weight` for Logistic Regression and Random Forest, and `scale_pos_weight` for XGBoost.

Synthetic oversampling such as SMOTE was not used in the final pipeline because the main concern appears to be source shift and mixed-type feature instability rather than only minority-class scarcity.

Payment strategies tested:

- **learned_imputation:** similar-employee payment imputer fitted within each training fold.
- **native_missing:** XGBoost receives numeric missingness directly where possible.
- **no_payment:** payment and salary features, including payment histories, are removed.

Model selection was based on validation PR-AUC. File3 was used only after model and threshold selection.

6 Model Results

Table 4 summarizes the final model comparison. The selected model is not necessarily the candidate with the single best file3 score; it is the model chosen before looking at file3. This preserves file3 as an external test set rather than an informal tuning set.

Table 4: Validation performance used for model selection.

Candidate	AUC	PR-AUC	F1	Precision	Recall
XGBoost__learned_imputation	0.7615	0.2052	0.2897	0.2027	0.5072
Random Forest__no_payment	0.7361	0.1949	0.2466	0.2072	0.3043
XGBoost__native_missing	0.7593	0.1942	0.2780	0.1895	0.5217
XGBoost__no_payment	0.7623	0.1925	0.2818	0.1805	0.6425
Random Forest__learned_imputation	0.7395	0.1876	0.2594	0.1848	0.4348
Logistic Regression__no_payment	0.7566	0.1860	0.3140	0.2387	0.4589
Logistic Regression__learned_imputation	0.7668	0.1859	0.2990	0.2112	0.5121

Table 5: External file3 performance after validation-based model selection.

Candidate	AUC	PR-AUC	F1	Precision	Recall	Recall@20%	Precision@20%
XGBoost__learned_imputation	0.7758	0.1462	0.1913	0.1147	0.5752	0.5163	0.1258
Random Forest__no_payment	0.7442	0.1373	0.1695	0.1493	0.1961	0.4510	0.1099
XGBoost__native_missing	0.7625	0.1254	0.2015	0.1237	0.5425	0.5033	0.1226
XGBoost__no_payment	0.7484	0.1057	0.1703	0.0969	0.7059	0.4444	0.1083
Random Forest__learned_imputation	0.7444	0.1736	0.2017	0.1350	0.3987	0.4444	0.1083
Logistic Regression__no_payment	0.7508	0.1109	0.1562	0.0866	0.7974	0.4837	0.1178
Logistic Regression__learned_imputation	0.7265	0.1047	0.1416	0.0771	0.8627	0.4575	0.1115

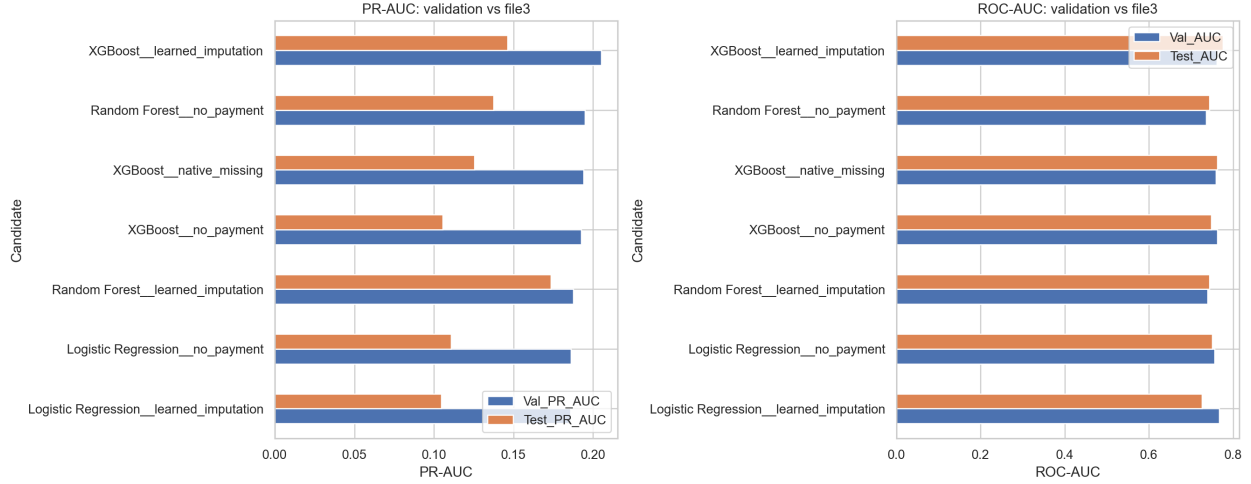


Figure 1: Validation and file3 performance comparison across final candidate models.

The figure compares candidate models using validation and external-test metrics. The selected XGBoost learned-imputation model was chosen by validation PR-AUC. Random Forest with learned imputation has a stronger file3 PR-AUC, but selecting it after seeing file3 would be test-set selection.

7 Feature Importance for the Selected Model

Interpretability was performed on the locked XGBoost learned-imputation model. Multiple importance methods were used because each answers a different question:

- Native XGBoost gain describes how the trees split.
- SHAP describes average contribution magnitude.
- File3 permutation measures external reliance by shuffling feature groups within period.
- Drop-column validation checks whether removing a feature hurts validation performance.

These methods are predictive explanations, not causal estimates.

Table 6: Top consensus features for the selected model.

Rank	Feature	Domain	Actionability	Consensus
1	avg_Payment	Compensation	Actionable	0.894
2	Yishuv	Location / commute	Descriptive	0.885
3	salary_change_hist_pre	Compensation	Actionable	0.849
4	age	Age / tenure	Descriptive	0.837
5	avg_Payment_hist_slope	Compensation	Actionable	0.822
6	tenure_years	Age / tenure	Descriptive	0.817
7	avg_illness_hist_vs_pre	Adverse event / wellness	Partially actionable	0.816

Rank	Feature	Domain	Actionability	Consensus
8	illness_cv	Absence / wellness	Partially action-able	0.814
9	stdevp_omes	Workload	Actionable	0.810
10	contract_type	Role / manager / organization	Partially action-able	0.808
11	manager_Code	Role / manager / organization	Partially action-able	0.802
12	vetek_months	Age / tenure	Descriptive	0.797
13	months_since_previous_Record	Record maturity / timing	Descriptive	0.796
14	stdevp_Payment	Compensation	Actionable	0.787
15	manager_Code_hist_prev	Role / manager / organization	Partially action-able	0.779

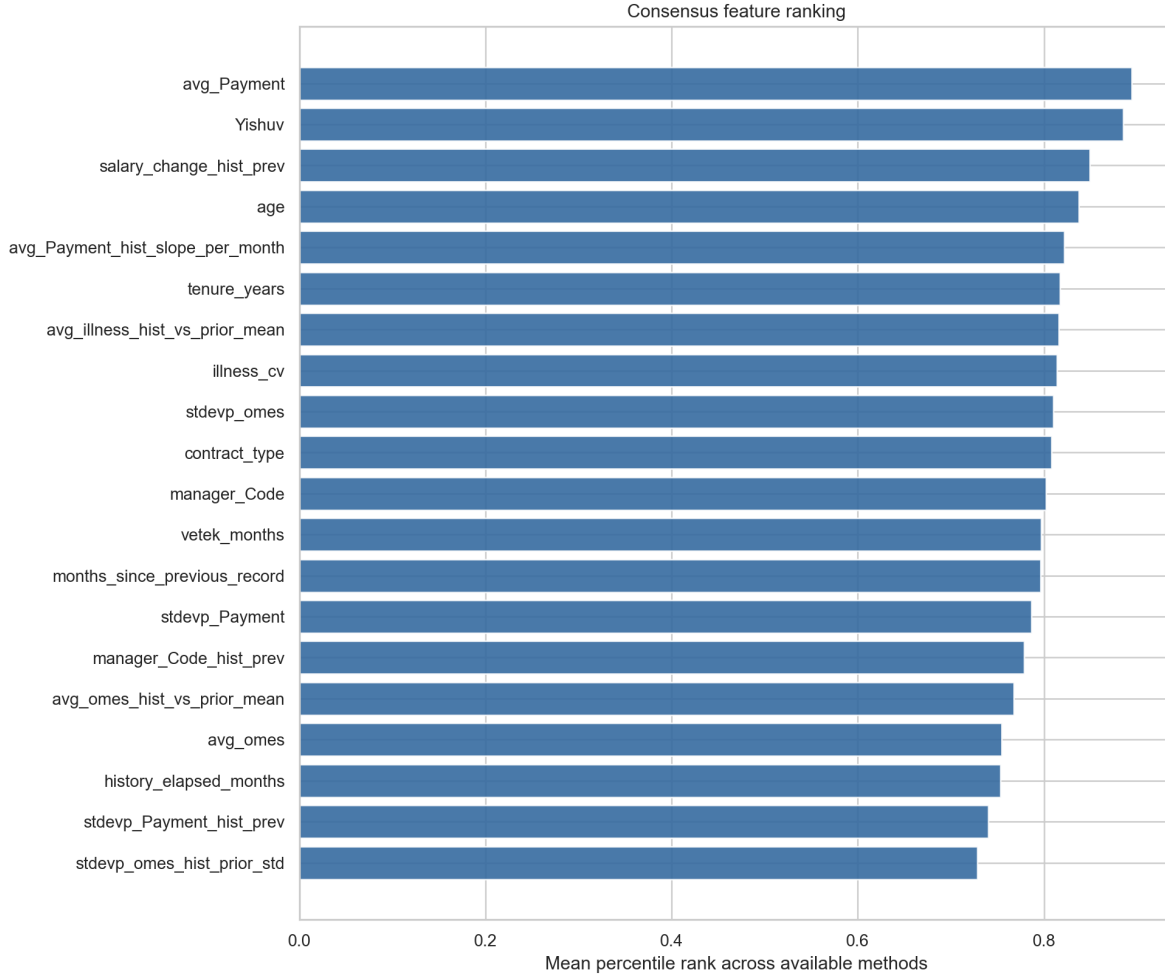


Figure 2: Consensus feature ranking across available importance methods.

This graph aggregates native gain, SHAP, permutation, and drop-column evidence where available.

It shows payment, age/tenure, salary-change history, location, illness, workload, and role/contract context in the selected model. Because not every method is available for every feature, the consensus score should be interpreted as a ranking aid rather than a formal statistical test.

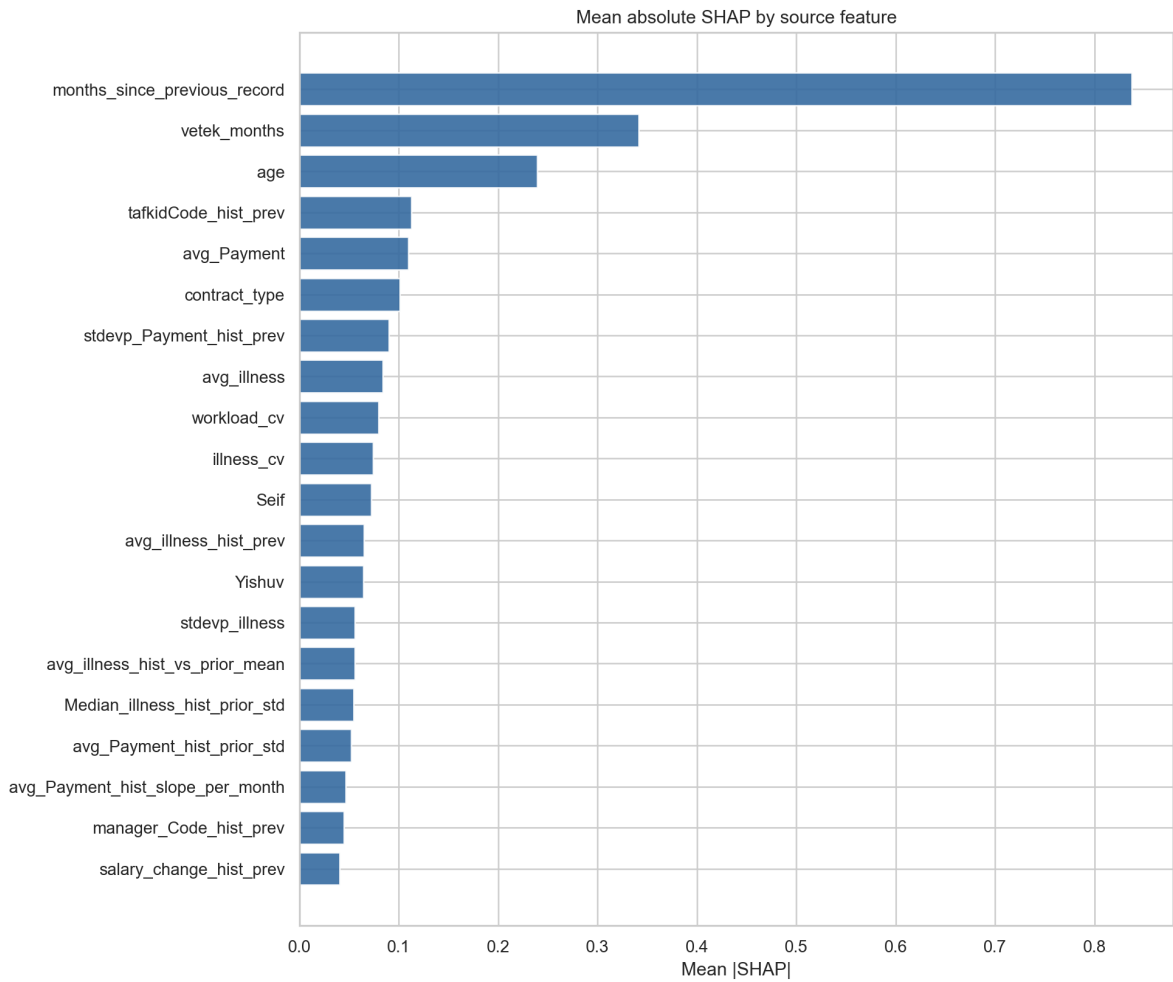


Figure 3: Mean absolute SHAP importance by source feature.

SHAP highlights months since previous record, tenure, and age as dominant descriptive signals. Compensation, contract type, illness, workload, and role-history features remain visible below those broad timing and life-cycle effects.

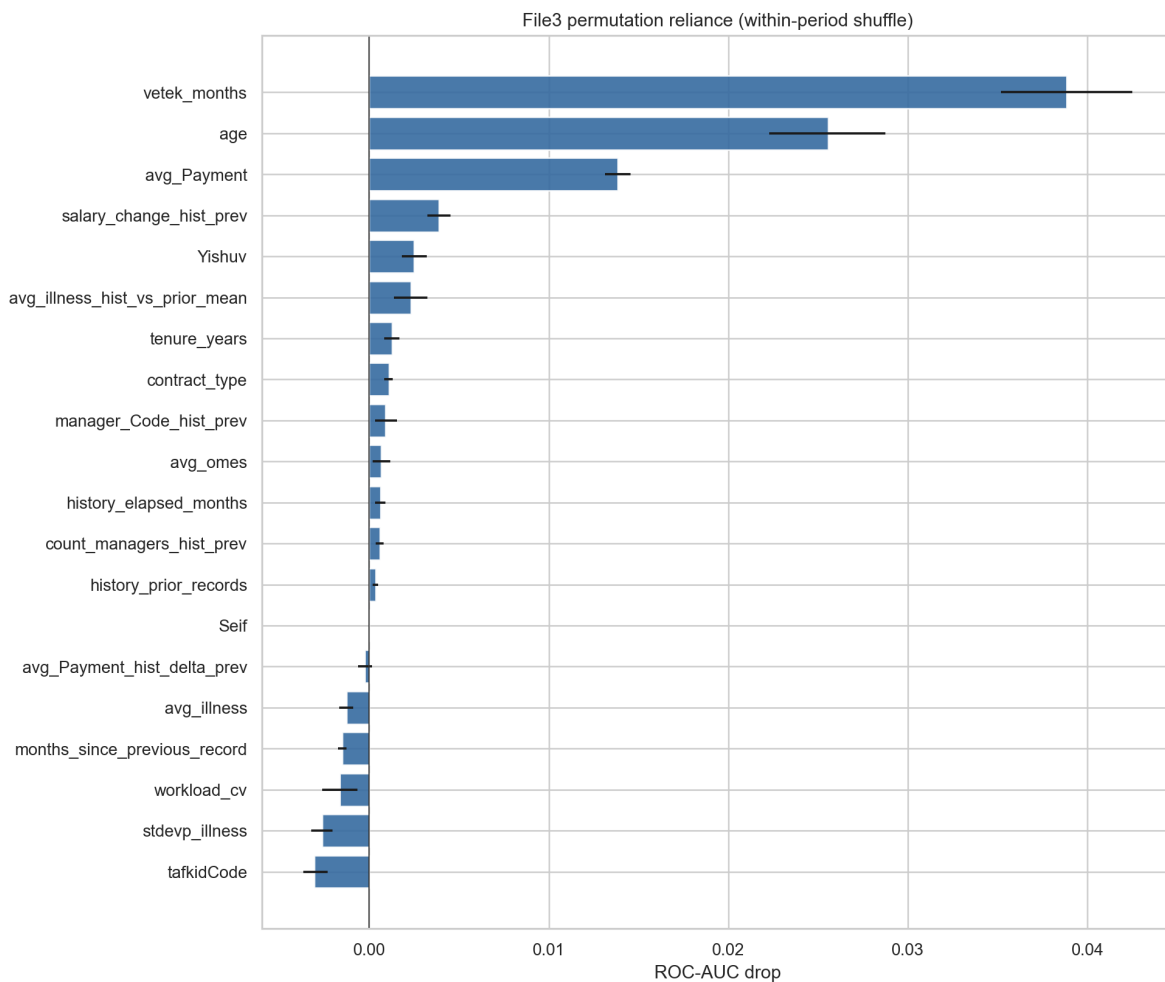


Figure 4: File3 permutation reliance using within-period shuffling.

The strongest external reliance is on tenure, age, and average payment. A positive AUC drop means performance decreased when that feature group was shuffled. Negative or near-zero values do not necessarily mean the feature is harmful; they may reflect noise, limited signal, or feature substitution by correlated variables.

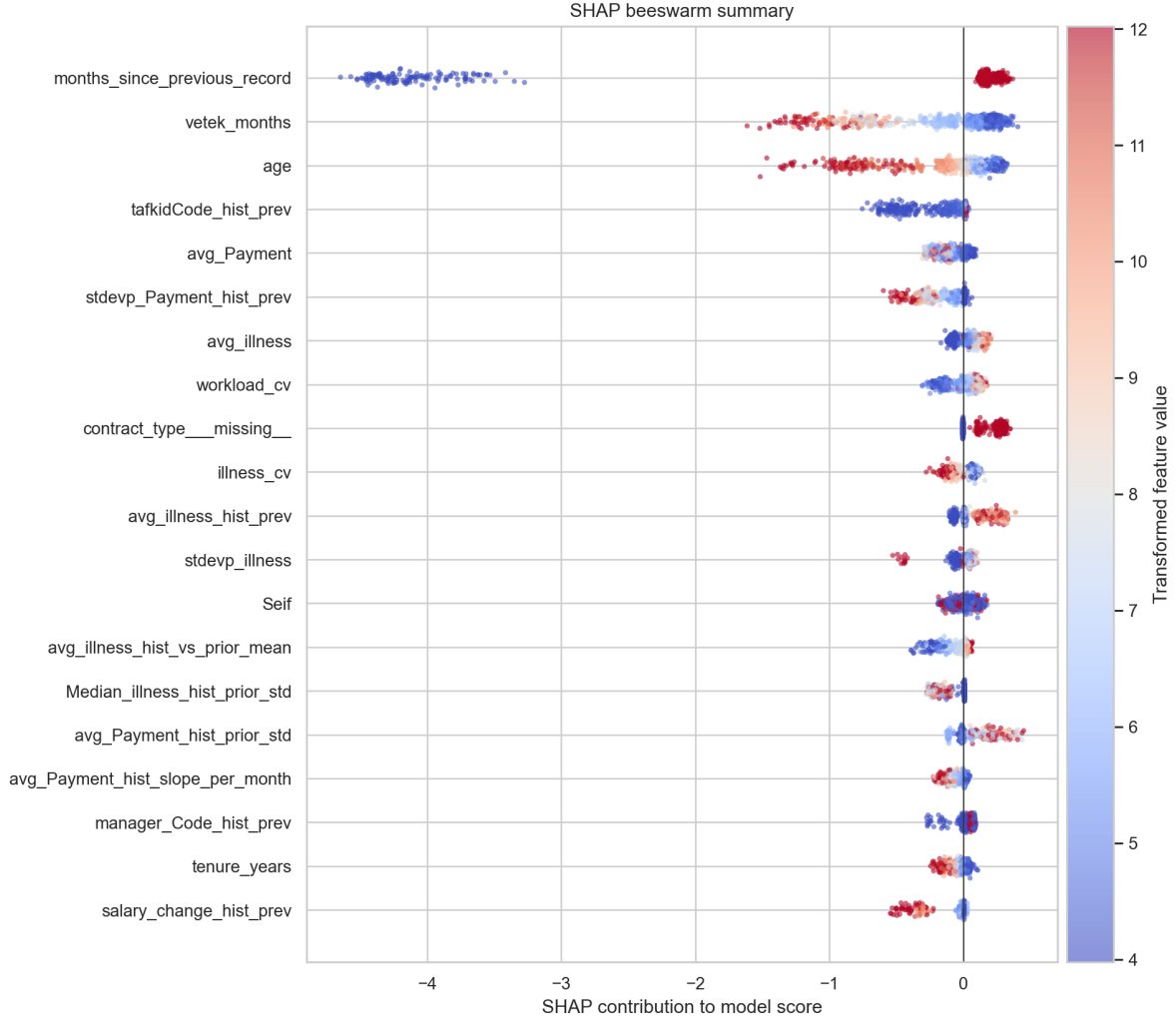


Figure 5: SHAP beeswarm for the selected XGBoost model.

Each point is a scored record; horizontal position indicates whether the feature pushed risk upward or downward. Dense overlap and encoded categorical features make the figure best suited for pattern recognition rather than exact measurement.

7.1 Domain and actionability summary

Feature importance was grouped into business domains to avoid over-interpreting individual correlated variables. Several payment-history features describe overlapping aspects of compensation trajectory, and several illness features describe overlapping absence patterns.

Table 7: SHAP importance by domain.

Domain	Actionability	Mean abs SHAP	Share
Data maturity / timing	Descriptive	0.898	22.8%
Compensation	Actionable	0.687	17.5%
Age / tenure	Descriptive	0.654	16.6%
Absence / wellness	Partially action- able	0.536	13.6%
Workload	Actionable	0.535	13.6%
Role / manager / organization	Partially action- able	0.496	12.6%
Location / commute	Descriptive	0.081	2.1%
Data availability	Descriptive	0.045	1.1%

Table 8: SHAP importance by actionability.

Actionability	Mean abs SHAP	Share
Descriptive	1.678	42.7%
Actionable	1.221	31.1%
Partially actionable	1.032	26.3%

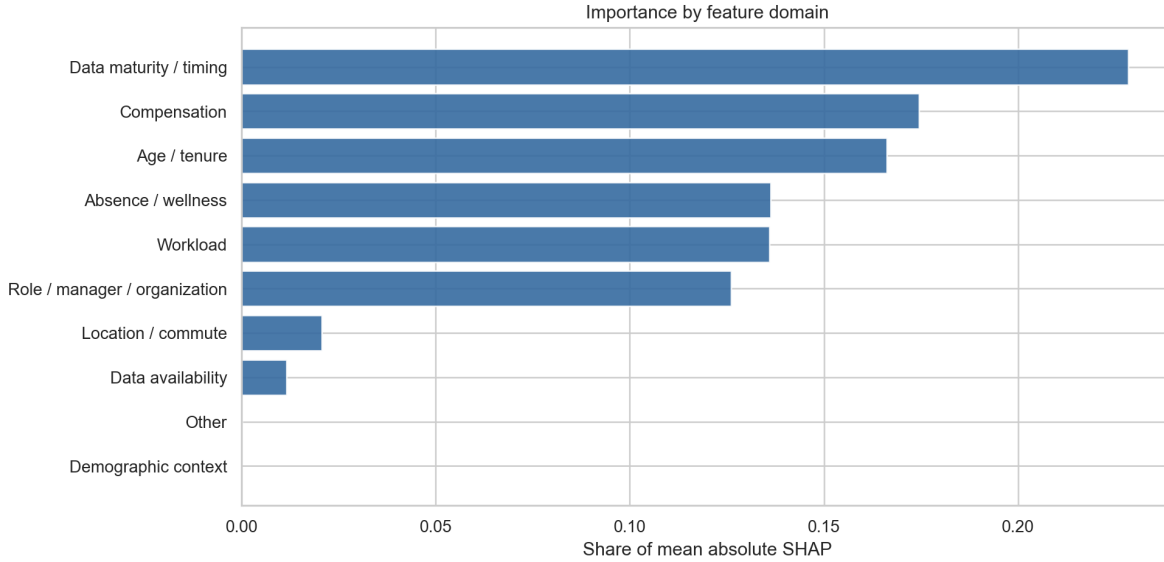


Figure 6: SHAP importance grouped by feature domain.

Data maturity/timing and age/tenure account for a large share of model behavior. Actionable areas remain meaningful: compensation, workload, illness/wellness, and role/manager/organization together represent a substantial share of model signal.

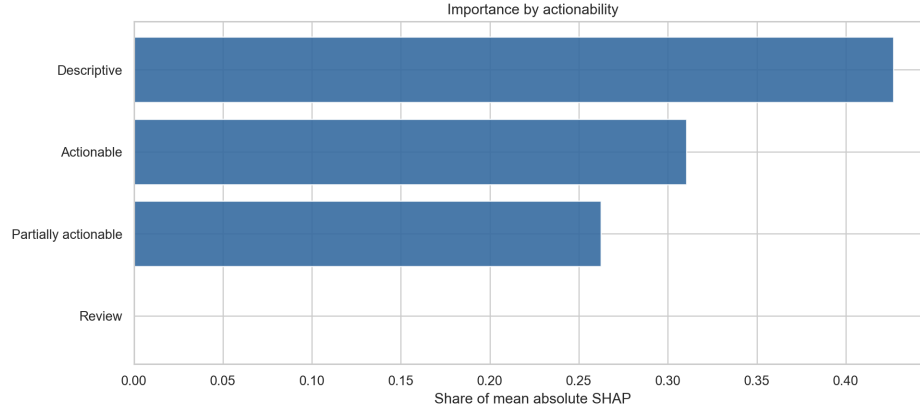


Figure 7: SHAP importance grouped by actionability.

Approximately 31% of SHAP importance is from directly actionable features and about 26% from partially actionable features. Descriptive variables are useful for segmentation but should not be treated as management levers.

8 Actionable Importance Within Age and Tenure Groups

Because age and tenure are strong descriptive predictors, a separate grouped analysis asked a more practical question: within comparable age or tenure groups, which actionable or partially actionable features still matter? This analysis used untouched file3 records, ranked actionable features by within-group SHAP, and measured within-group permutation PR-AUC drops when enough positives and negatives were available.

8.1 Within age bands

Table 9: Top actionable SHAP features within age bands.

Age band	Rows	Pos.	Prev.	Rank	Feature	Domain	SHAP
18-34	721	76	10.5%	1	avg_Payment	Compensation	0.123
18-34	721	76	10.5%	2	contract_type	Role / manager / organization	0.115
18-34	721	76	10.5%	3	tafkidCode_hist_prev	Role / manager / organization	0.111
18-34	721	76	10.5%	4	avg_illness	Absence / well-ness	0.089
35-49	1,360	60	4.4%	1	stdevp_Payment_hist_prev	Compensation	0.109
35-49	1,360	60	4.4%	2	avg_Payment	Compensation	0.107
35-49	1,360	60	4.4%	3	tafkidCode_hist_prev	Role / manager / organization	0.105
35-49	1,360	60	4.4%	4	contract_type	Role / manager / organization	0.102

Age band	Rows	Pos.	Prev.	Rank	Feature	Domain	SHAP
50-59	710	14	2.0%	1	stdevp_Payment_hist_prev	Compensation	0.122
50-59	710	14	2.0%	2	tafkidCode_hist_prev	Role / manager / organization	0.108
50-59	710	14	2.0%	3	avg_Payment	Compensation	0.100
50-59	710	14	2.0%	4	contract_type	Role / manager / organization	0.083
60+	349	3	0.9%	1	tafkidCode_hist_prev	Role / manager / organization	0.159
60+	349	3	0.9%	2	contract_type	Role / manager / organization	0.099
60+	349	3	0.9%	3	avg_Payment	Compensation	0.099
60+	349	3	0.9%	4	illness_cv	Absence / well-ness	0.083

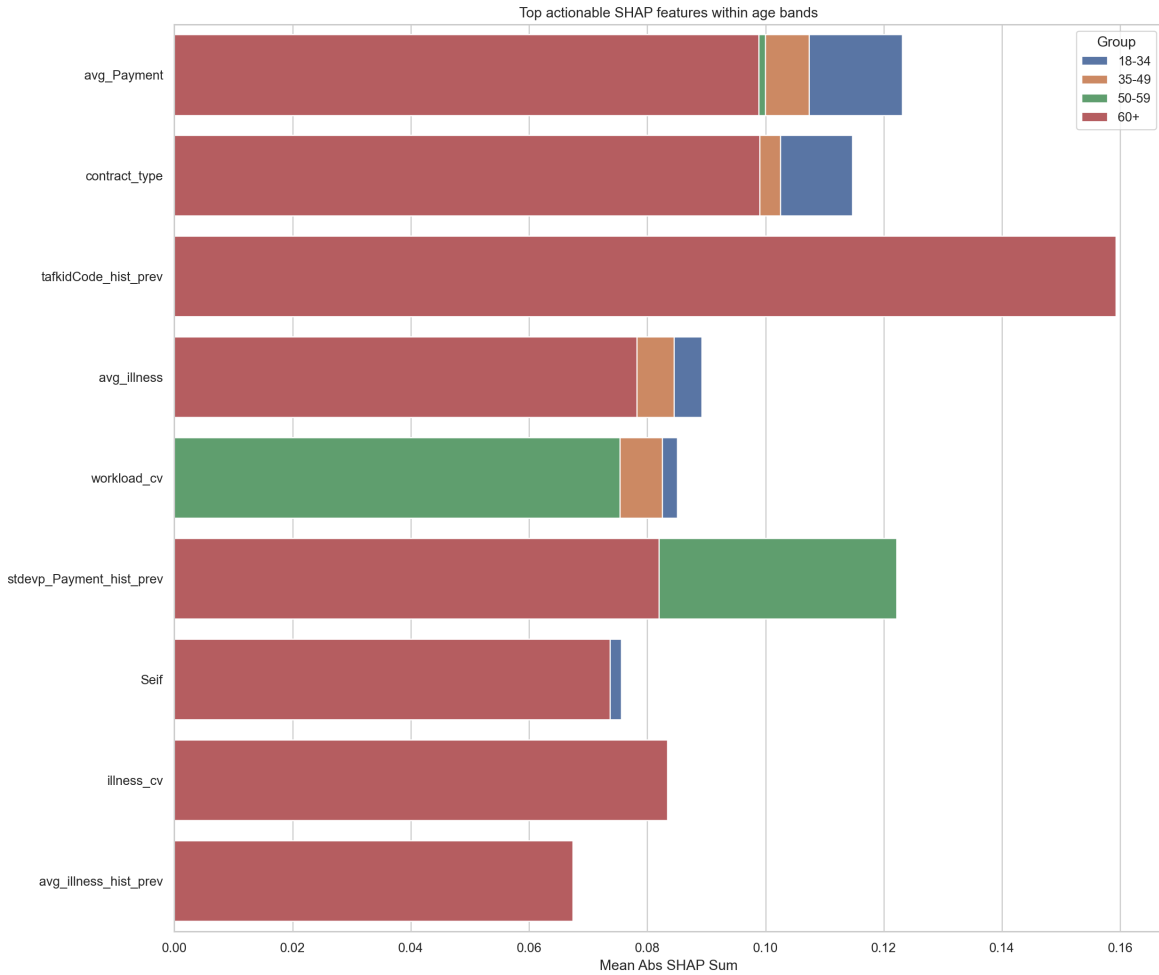


Figure 8: Top actionable SHAP features within age bands.

Average payment, payment variability, contract type, role history, illness, and workload remain

visible after segmenting by age. The 60+ group has only three positives, so its ranking is descriptive and low-confidence.

Table 10: Top actionable permutation PR-AUC drops within age bands.

Age band	Feature	Pos.	Prev.	Base PR-AUC	PR-AUC drop
18-34	avg_Payment	76	10.5%	0.2135	0.0379
18-34	illness_cv	76	10.5%	0.2135	0.0059
18-34	avg_Payment_hist_prior_std	76	10.5%	0.2135	0.0050
35-49	avg_Payment	60	4.4%	0.0846	0.0087
35-49	stdevp_Payment_hist_prev	60	4.4%	0.0846	0.0037
35-49	avg_illness_hist_vs_prior_mean	60	4.4%	0.0846	0.0031
50-59	avg_Payment	14	2.0%	0.0769	0.0157
50-59	illness_cv	14	2.0%	0.0769	0.0124
50-59	avg_Payment_hist_prior_std	14	2.0%	0.0769	0.0035

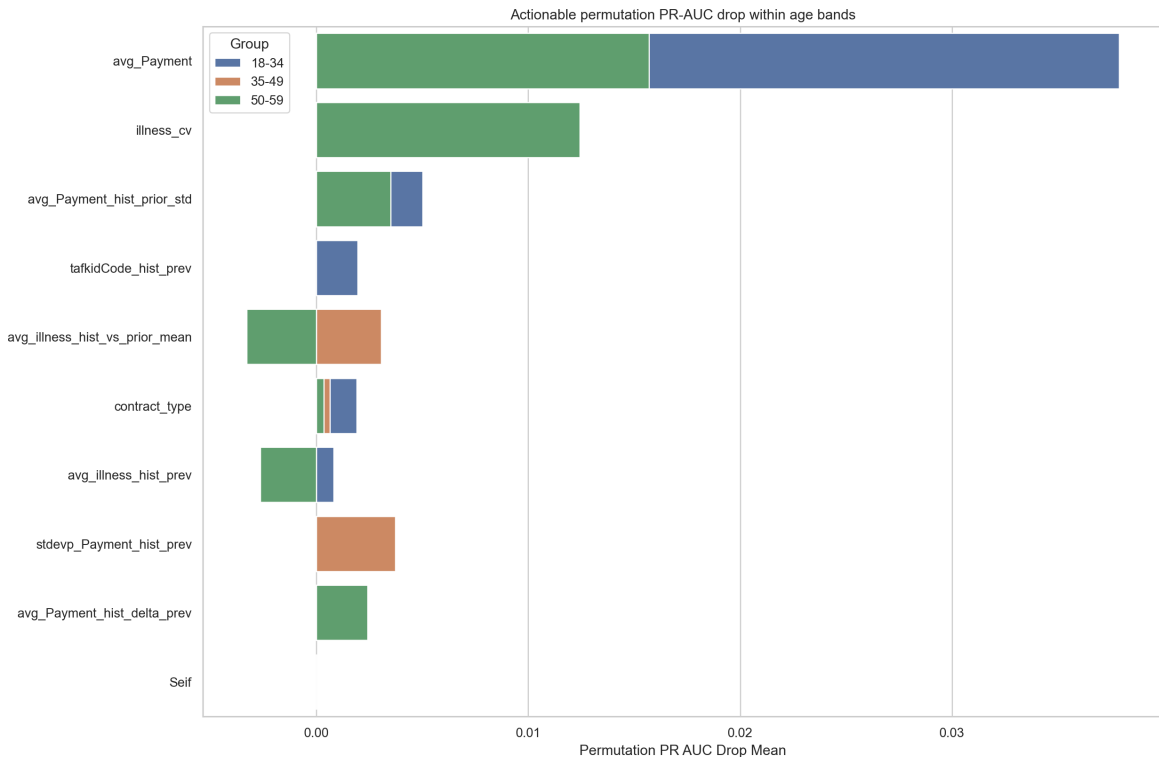


Figure 9: Within-age-band actionable permutation PR-AUC drops.

Average payment is the clearest actionable reliance signal in the 18-34, 35-49, and 50-59 age bands. The 60+ band is omitted from permutation interpretation because it had too few positives.

8.2 Within tenure bands

Table 11: Top actionable SHAP features within tenure bands.

Tenure	Rows	Pos.	Prev.	Rank	Feature	Domain	SHAP
< 2y	775	77	9.9%	1	avg_Payment	Compensation	0.142
< 2y	775	77	9.9%	2	contract_type	Role / manager / organization	0.107
< 2y	775	77	9.9%	3	workload_cv	Workload	0.094
< 2y	775	77	9.9%	4	avg_illness	Absence / well-ness	0.078
2-5y	749	39	5.2%	1	avg_Payment	Compensation	0.116
2-5y	749	39	5.2%	2	stdevp_Payment_hist_prev	Compensation	0.115
2-5y	749	39	5.2%	3	tafkidCode_hist_prev	Role / manager / organization	0.115
2-5y	749	39	5.2%	4	contract_type	Role / manager / organization	0.108
5-10y	795	24	3.0%	1	stdevp_Payment_hist_prev	Compensation	0.146
5-10y	795	24	3.0%	2	tafkidCode_hist_prev	Role / manager / organization	0.141
5-10y	795	24	3.0%	3	avg_Payment_hist_prior	Compensation	0.096
5-10y	795	24	3.0%	4	contract_type	Role / manager / organization	0.094
10y+	821	13	1.6%	1	stdevp_Payment_hist_prev	Compensation	0.125
10y+	821	13	1.6%	2	tafkidCode_hist_prev	Role / manager / organization	0.123
10y+	821	13	1.6%	3	contract_type	Role / manager / organization	0.094
10y+	821	13	1.6%	4	avg_Payment_hist_delta	Compensation	0.092

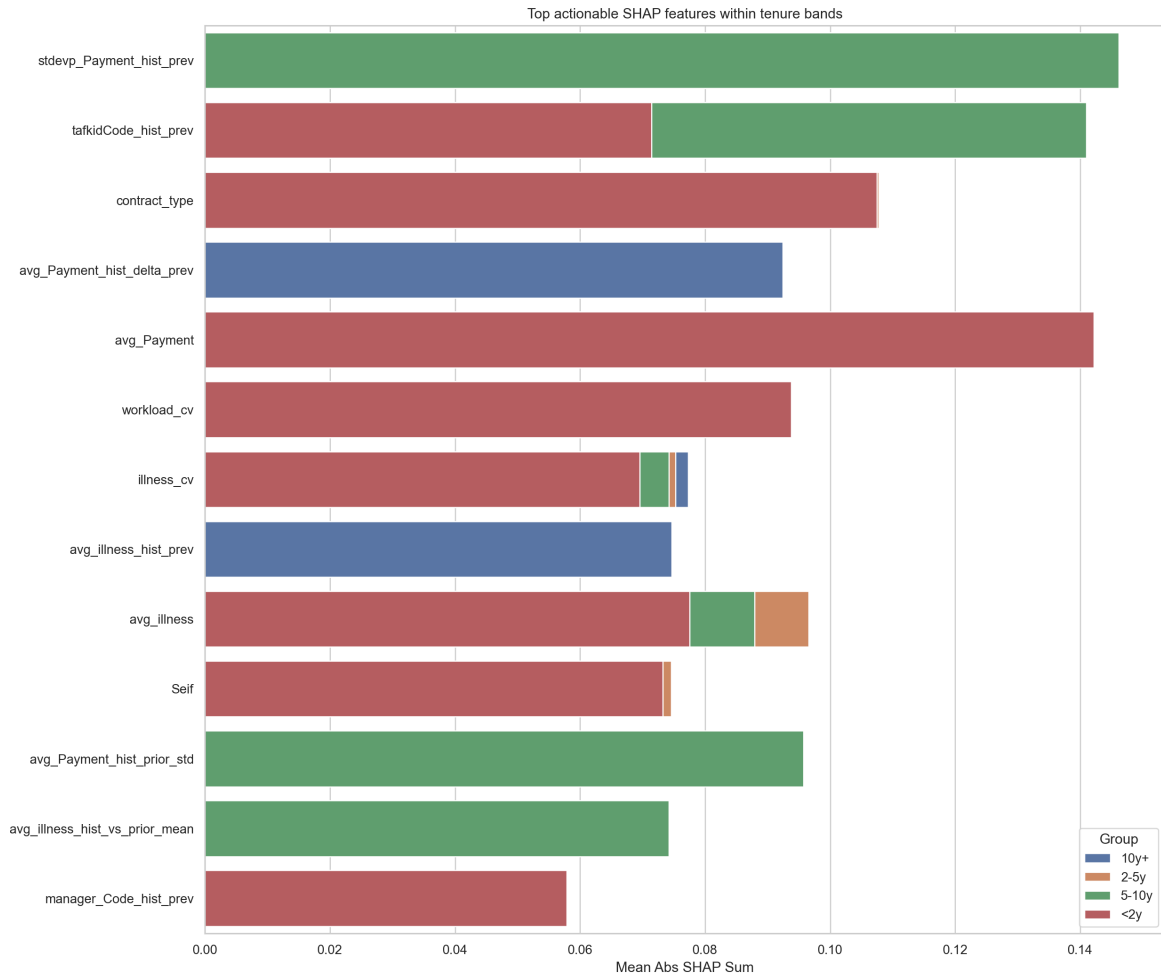


Figure 10: Top actionable SHAP features within tenure bands.

For employees with less than two years tenure, average payment is the leading actionable feature. For 5-10 and 10+ years tenure, payment variability/history, role history, illness history, and contract type become more prominent.

Table 12: Top actionable permutation PR-AUC drops within tenure bands.

Tenure	Feature	Pos.	Prev.	Base PR-AUC	PR-AUC drop
< 2y	avg_Payment	77	9.9%	0.2294	0.0488
< 2y	illness_cv	77	9.9%	0.2294	0.0029
< 2y	contract_type	77	9.9%	0.2294	0.0028
2-5y	illness_cv	39	5.2%	0.0761	0.0052
2-5y	avg_Payment	39	5.2%	0.0761	0.0036
2-5y	avg_Payment_hist_prior_std	39	5.2%	0.0761	0.0027
5-10y	avg_illness	24	3.0%	0.1418	0.0264
5-10y	avg_Payment	24	3.0%	0.1418	0.0247
5-10y	avg_Payment_hist_delta_prev	24	3.0%	0.1418	0.0063
10y+	avg_illness_hist_vs_prior_mean	13	1.6%	0.0759	0.0053
10y+	stdevp_Payment_hist_prev	13	1.6%	0.0759	0.0040
10y+	tafkidCode_hist_prev	13	1.6%	0.0759	0.0039

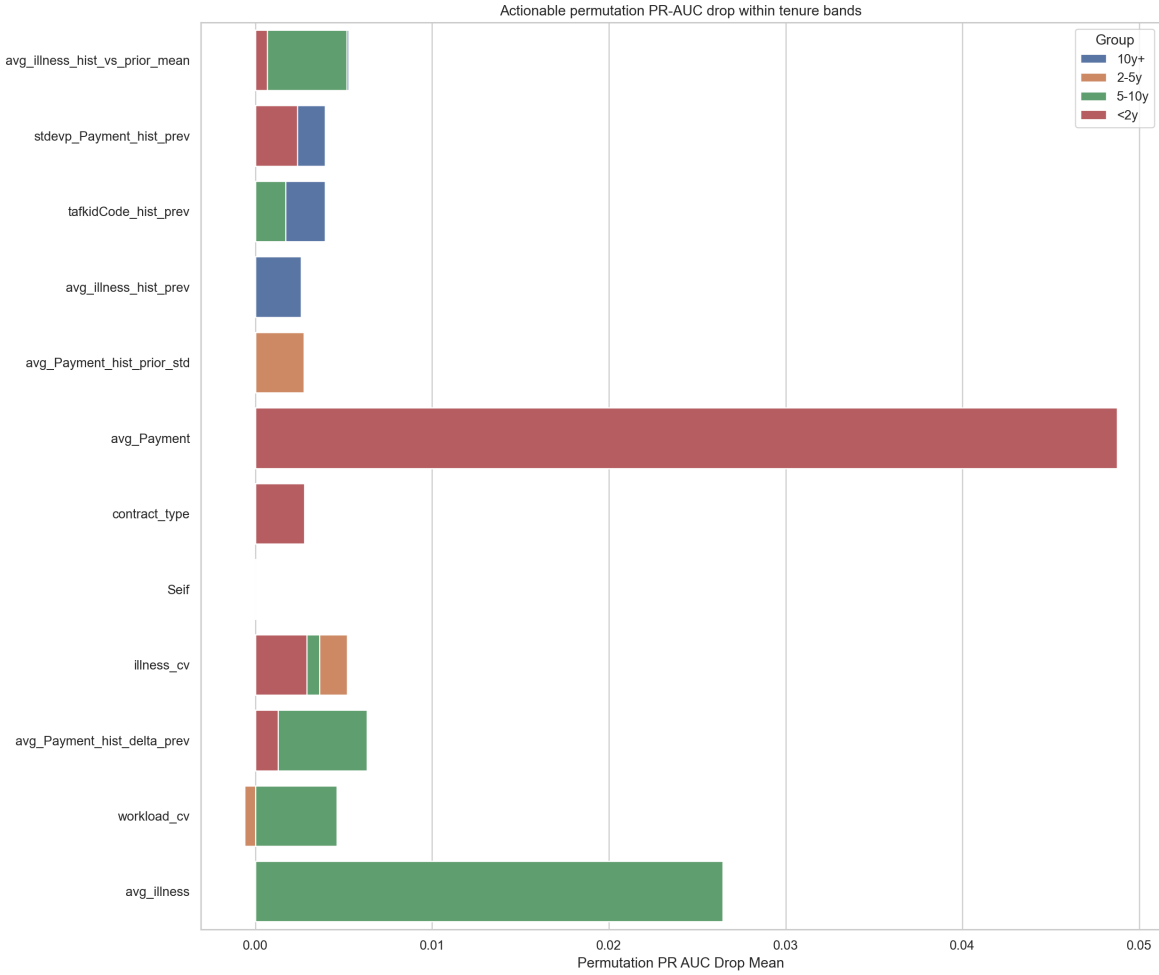


Figure 11: Within-tenure-band actionable permutation PR-AUC drops.

The permutation results suggest different levers by tenure stage: payment dominates among newer employees; illness and payment both matter in the 5-10 year band; and illness-history and payment-history features matter most for long-tenure employees. Some negative or near-zero drops in the full CSV should be treated as weak evidence rather than evidence of causal protection.

9 Streamlit Dashboard Integration

The final model was incorporated into the existing Streamlit executive dashboard while preserving the original visual style and tab structure. A new final-model dataset option builds dashboard-compatible latest-employee records from final file3 predictions. The micro-level what-if simulator was revised so that it reflects the final feature-importance results rather than merely showing percent input changes.

Dashboard changes:

- Macro and meso views continue to summarize turnover probability, risk categories, budget-section risk, tenure-risk patterns, and highest-risk employees.
- The micro view now supports the final artifact and scores prepared final-model rows using the locked XGBoost pipeline.
- What-if levers focus on salary/payment, workload, illness, contract type, job rank, and budget section.
- For final-model mode, the lever-impact chart reports estimated risk-point impact from each lever, aligning the interface more closely with model interpretability outputs.

10 Limitations and Recommended Next Steps

The model demonstrates useful ranking performance, but several limitations remain. These are methodological constraints imposed by the data and prediction problem rather than simply implementation defects.

- **Generalizability:** File3 differs from file1/file2 in prevalence and likely feature distributions. The model may learn source-specific patterns that transfer imperfectly.
- **Payment imputation:** File2 payment imputation is noisy. Payment is important to the final model, so systematic imputation error can affect both performance and explanations.
- **Low event prevalence:** File3 prevalence is only 4.87%, so precision remains modest even when ranking metrics are above baseline.
- **Small subgroups:** The 60+ age group contains only three positives in file3, making subgroup permutation PR-AUC unreliable.
- **Causality:** Feature importance is predictive, not causal. Compensation, workload, and illness features can guide review, but interventions should be validated with domain knowledge and policy constraints.

- **Temporal framing:** A future improvement would define a prediction horizon, such as resignation within the next 3, 6, or 12 months, rather than treating every row as an undifferentiated turnover prediction row.

Recommended next research step. The strongest next methodological improvement is to quantify source drift between file1/file2 and file3 and then test portable/actionable feature sets under source-aware validation. This would clarify whether weaker PR-AUC is caused by class imbalance, missingness, population shift, or unstable company-specific codes.

Additional useful extensions:

1. Reframe the target as resignation within a fixed future horizon, such as 3, 6, or 12 months.
2. Compare row-level predictions against latest-row-per-employee predictions, since HR interventions usually occur at the employee level.
3. Run train-file1/validate-file2 and train-file2/validate-file1 experiments to quantify cross-source portability.
4. Compare feature sets that remove company-specific codes, age, tenure, and location to determine the tradeoff between predictive performance and generalizability.
5. Evaluate calibration on file3, especially if probabilities are used directly in the dashboard rather than only for ranking.

A Files Produced by the Final Pipeline

- Model artifact: `artifacts/final_best_model.pkl`
- Predictions: `output/final/file3_predictions.xlsx`
- Model comparison: `output/final/model_comparison.csv`
- Feature-importance report: `output/final/feature_importance/feature_importance_report.txt`
- Grouped actionable report: `output/final/feature_importance/grouped_actionable/grouped_actionable_report.txt`
- Streamlit integration helpers: `src/final_dashboard.py` and updated `src/inference.py`

B Metric Definitions

- **ROC-AUC:** Measures how often the model ranks a positive example above a negative example. It is useful for ranking but can look optimistic under heavy class imbalance.
- **PR-AUC:** Summarizes precision-recall behavior and is more sensitive to rare-event prediction quality. It is the primary selection metric in this final pipeline.
- **F1:** Harmonic mean of precision and recall at a selected threshold.
- **Recall@Top20%:** Share of positives captured when reviewing the top 20% highest-risk rows.
- **Precision@Top20%:** Positive rate among the top 20% highest-risk rows.

- **Brier score:** Probability calibration error; lower is better.
- **Log loss:** Penalizes confident wrong probabilities; lower is better.